

DES(S-box & P-box).

Des

Data encryption standard

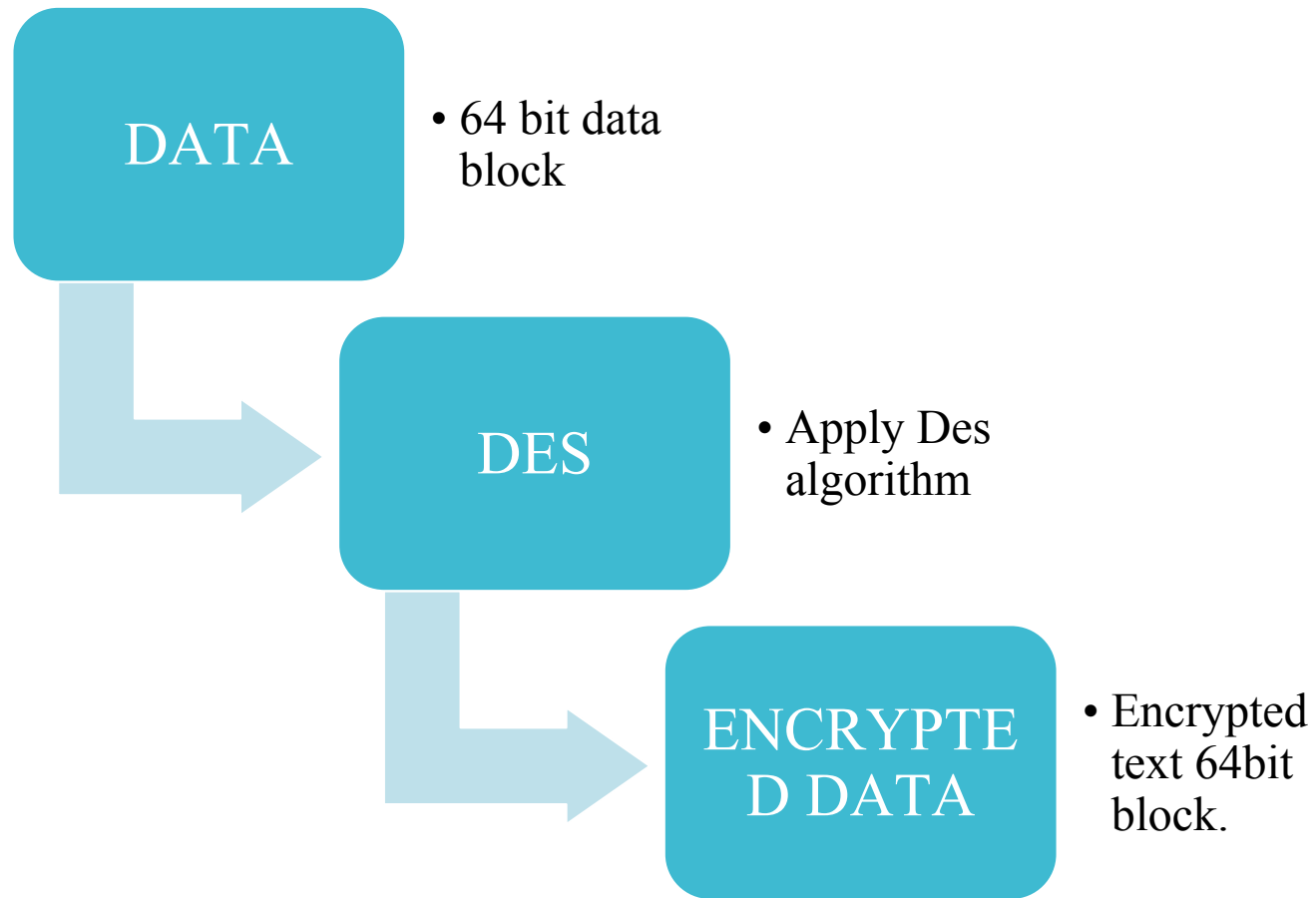
- Des has been a cryptographic algorithm over three decade
- Des has been found vulnerable against very powerful attack.
- It has been a landmark for cryptographic algorithm.

Aim to learn des

- Understand Des
- To dissect and understand real life problem.
- Useful for understanding other algorithm of cryptographic

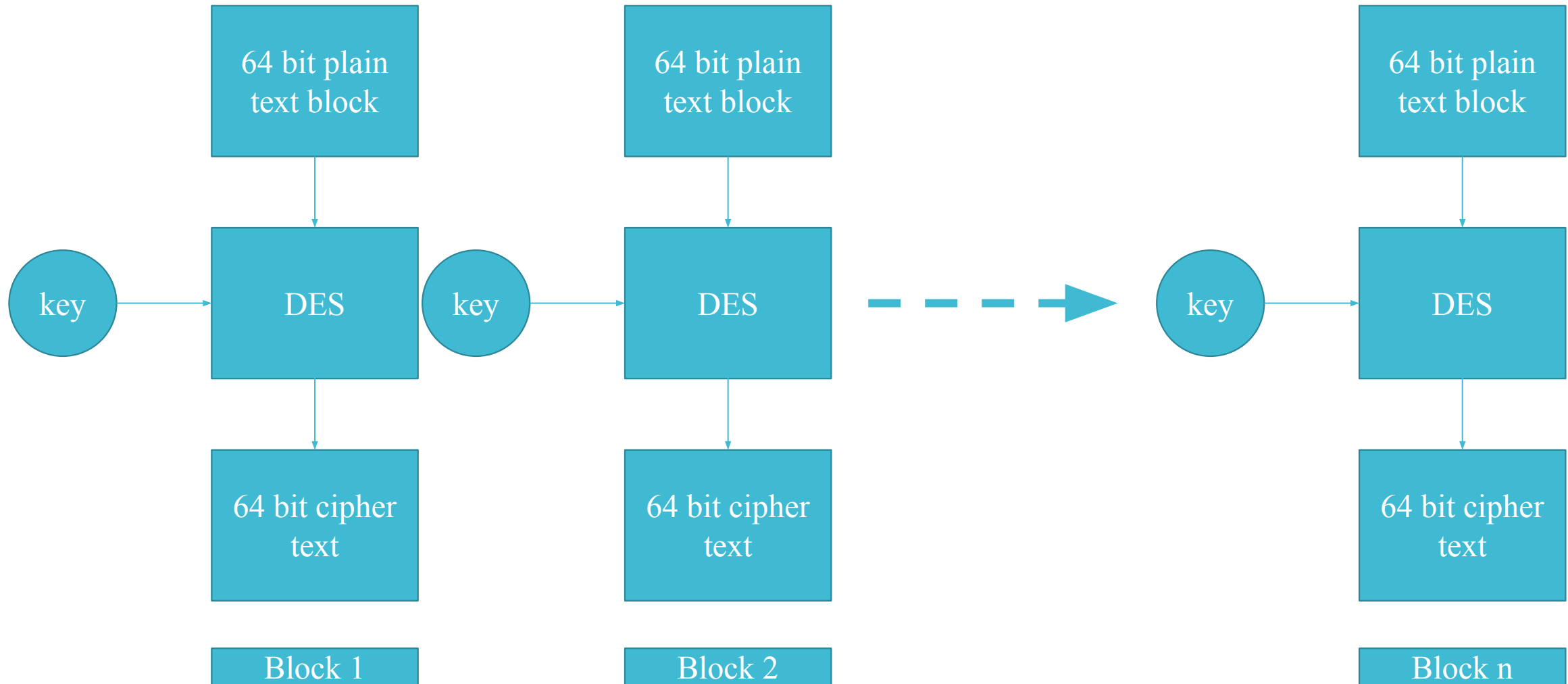
Des uses

- ECB Electronic Code block.
- CBC Cipher Block Chaining mode.
- CFB Cipher Feed Back

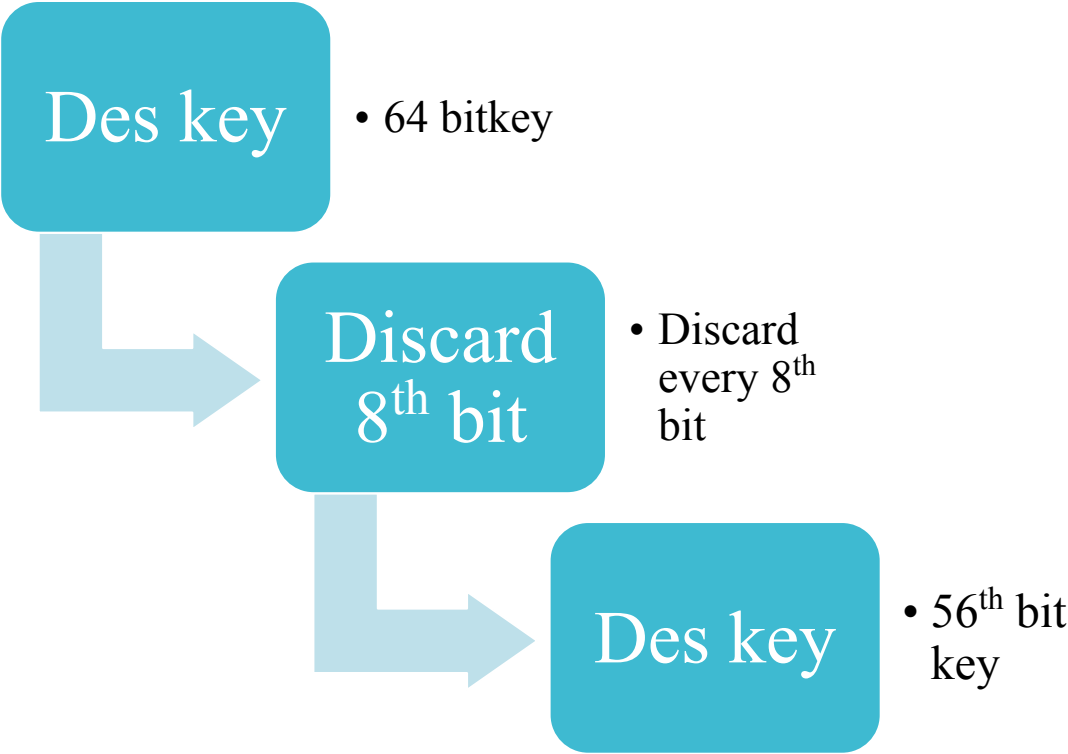


- Type :- BLOCK
- For encryption DES
- For decryption DES with minor difference.
- Key length used key is 56bit.

Conceptual working of Des



How we get 56 bit key.



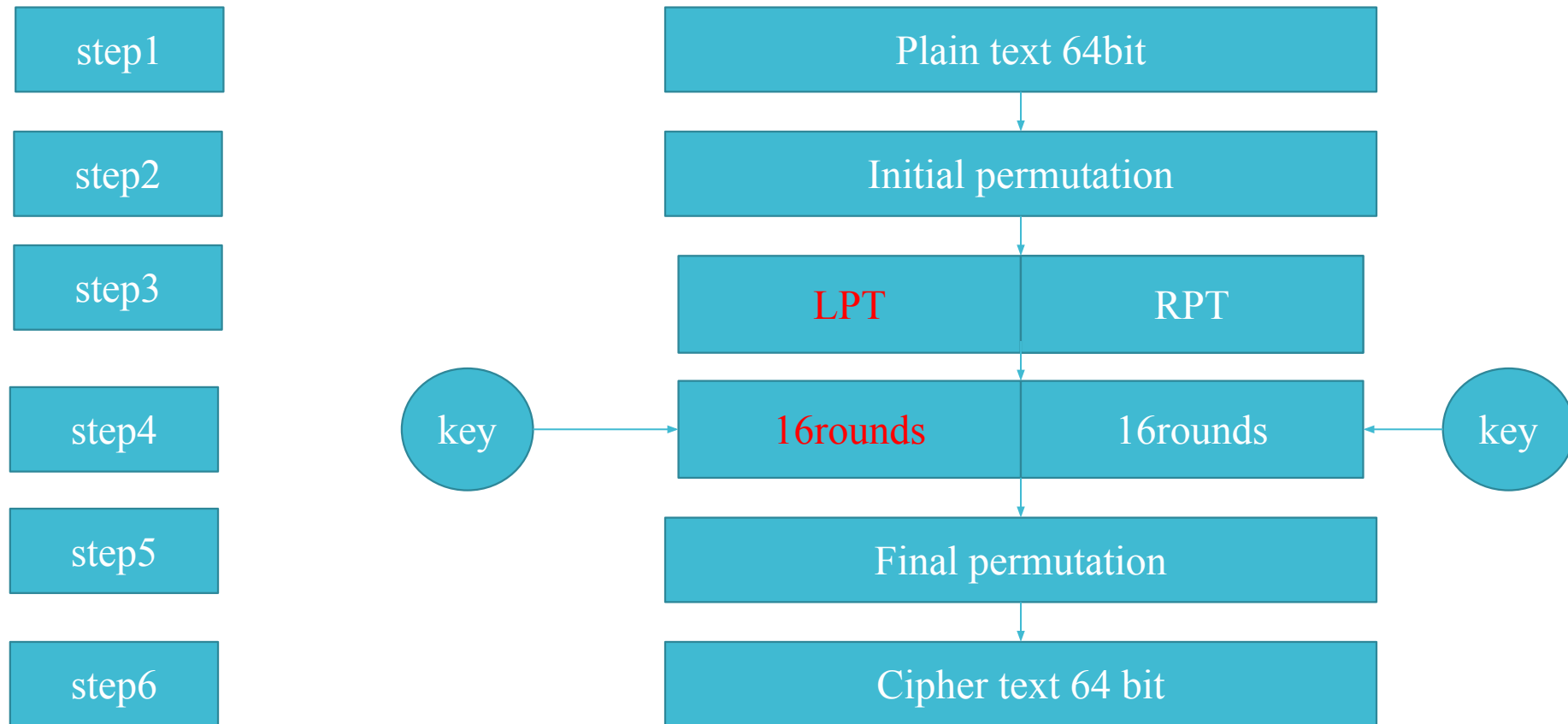
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64

Discarded key

Des is based
on two
fundamental

- Substitution (Confusion)
 - Transpositions (diffusion)
-
- Consist of 16 rounds
 - Each round contain
 - Substitution (Confusion)
 - Transpositions (diffusion)

Steps for Des algorithm



- Plain text is the simple text , information or msg to be send from sender to receiver.
- Plain text may be very much confidential to user thus provide an encryption technique to provide safety during transaction and storage format.

Step 2 Initial permutation.

Initial Data.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48
49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64



Initial Permutation.

58	50	42	34	26	18	10	2	60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6	64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1	59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5	63	55	47	39	31	23	15	7



58		4	62		7
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32bit	32bit
LPT	RPT

- Here conversion take place as follows.

1	58
2	50
3	40
4	34
5	26
6	18
7	10
8	2
9	60
.	
.	
.	
.	
64	7

- Always read from left to right
- Always read from top to bottom.

Step 3 16 Rounds

Key Transformation

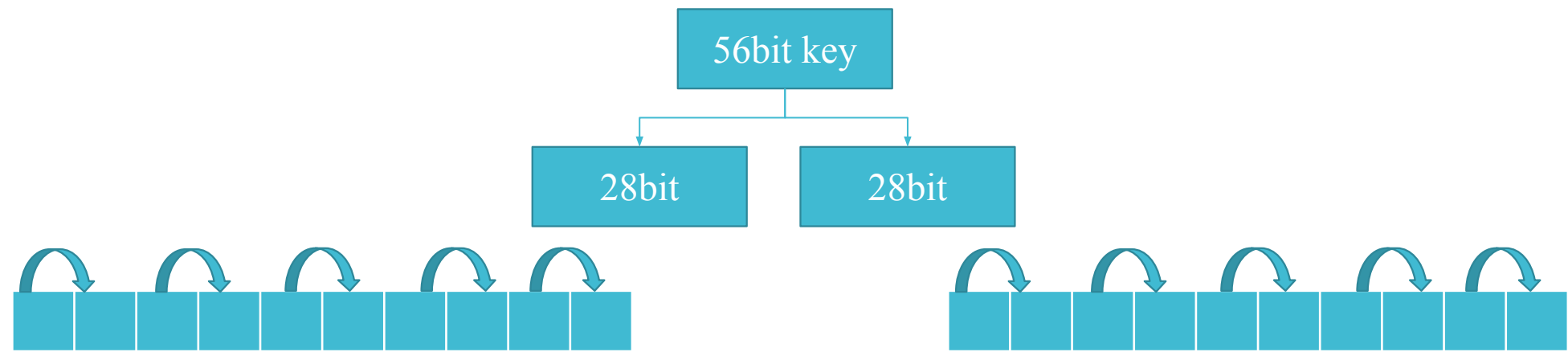
Expansion and Permutation

S-Box Substitution

P-box Substitution

XOR and SWAP

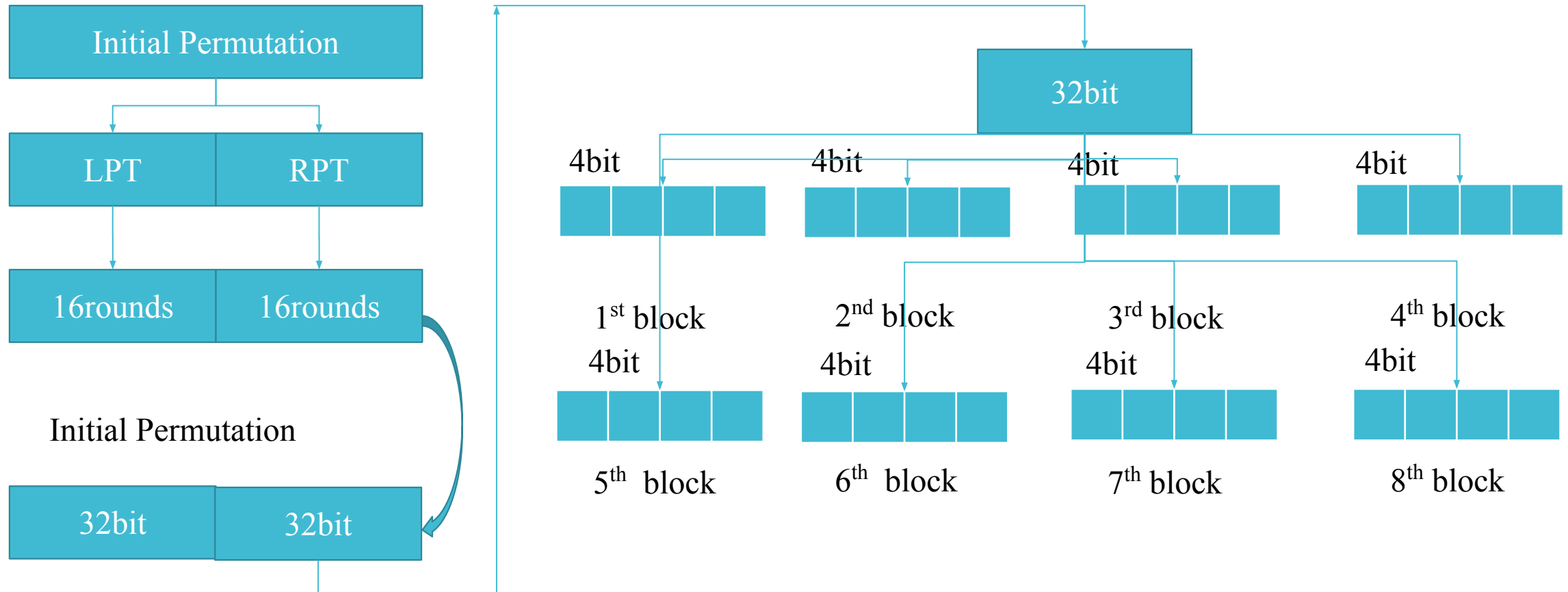
Step 1 of 3 key Transformation.



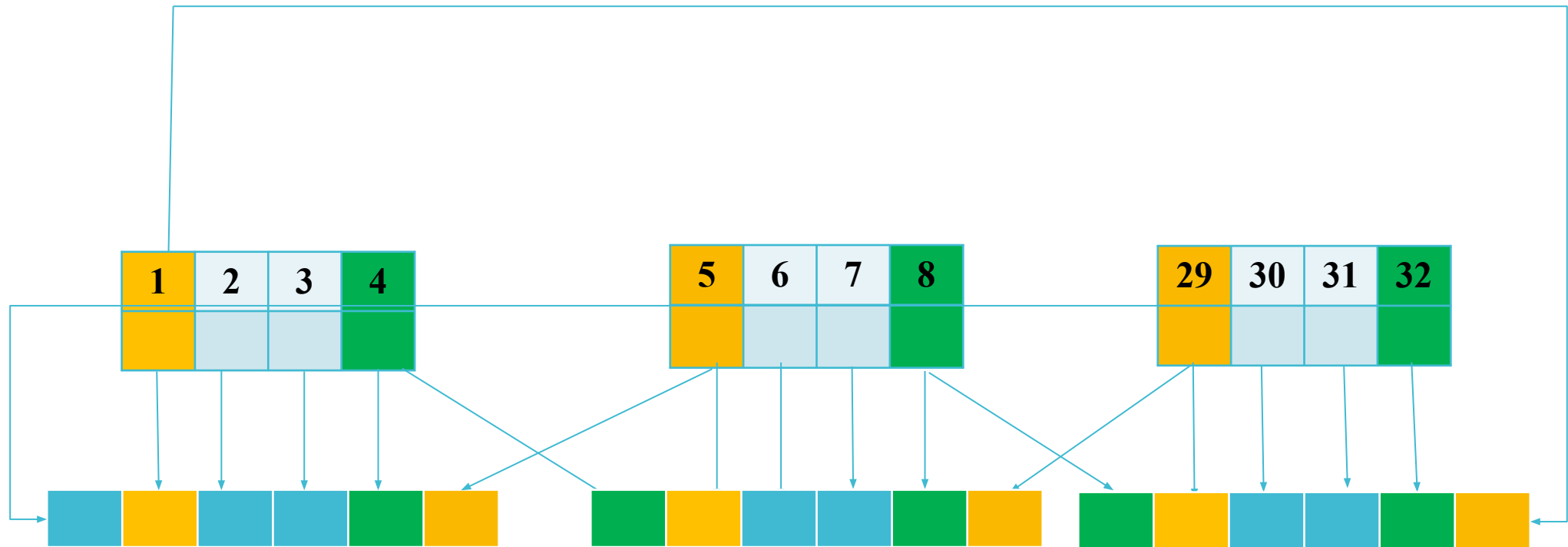
rounds	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
No of bits shifted	1	2	1	1	1	2	1	2	1	2	1	1	1	1	1	1

- For each round different subset of the key is used that make not easy to crack.
- Here in each of the round the bits are shifted randomly.

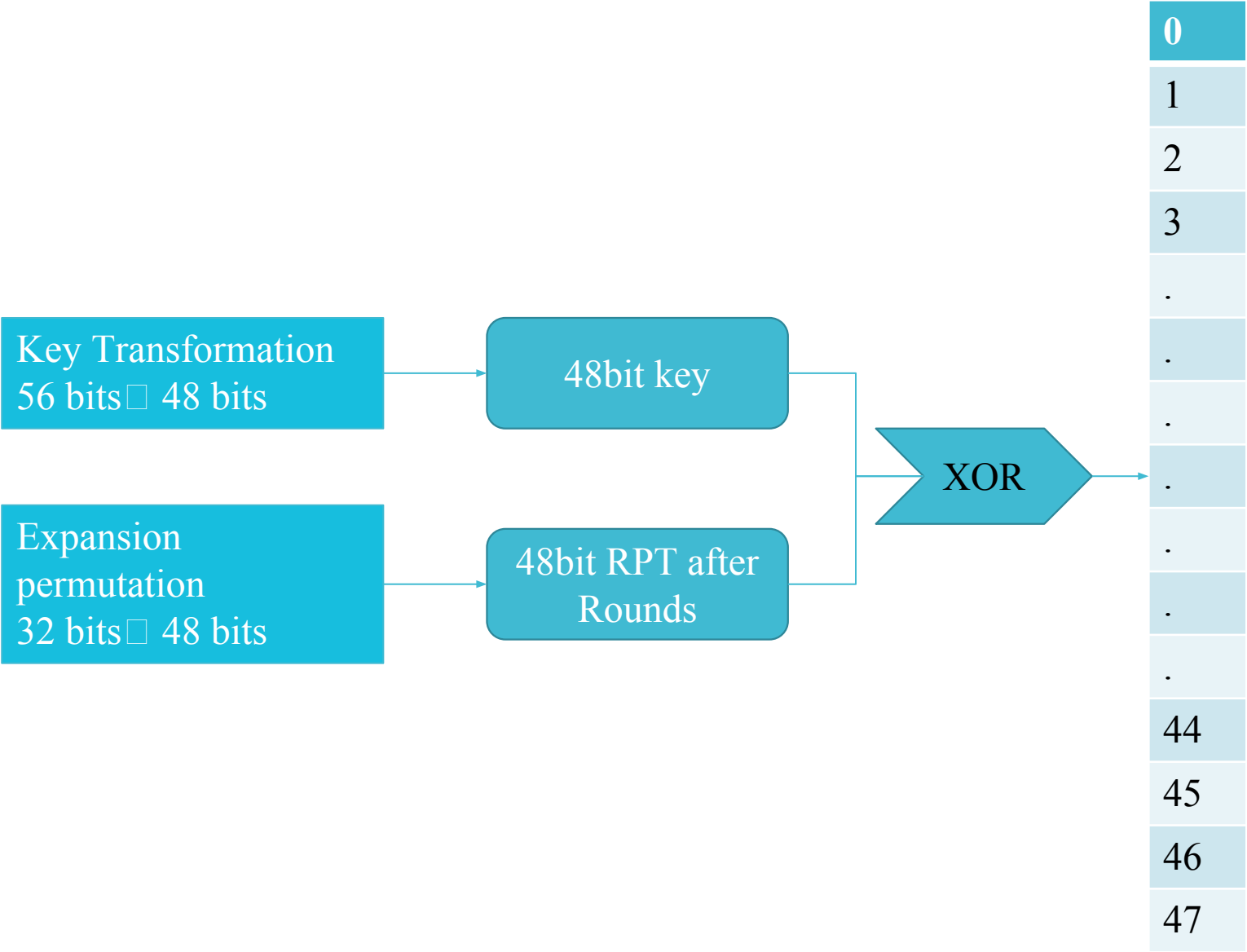
Step 2 of 3 Expansion Permutation

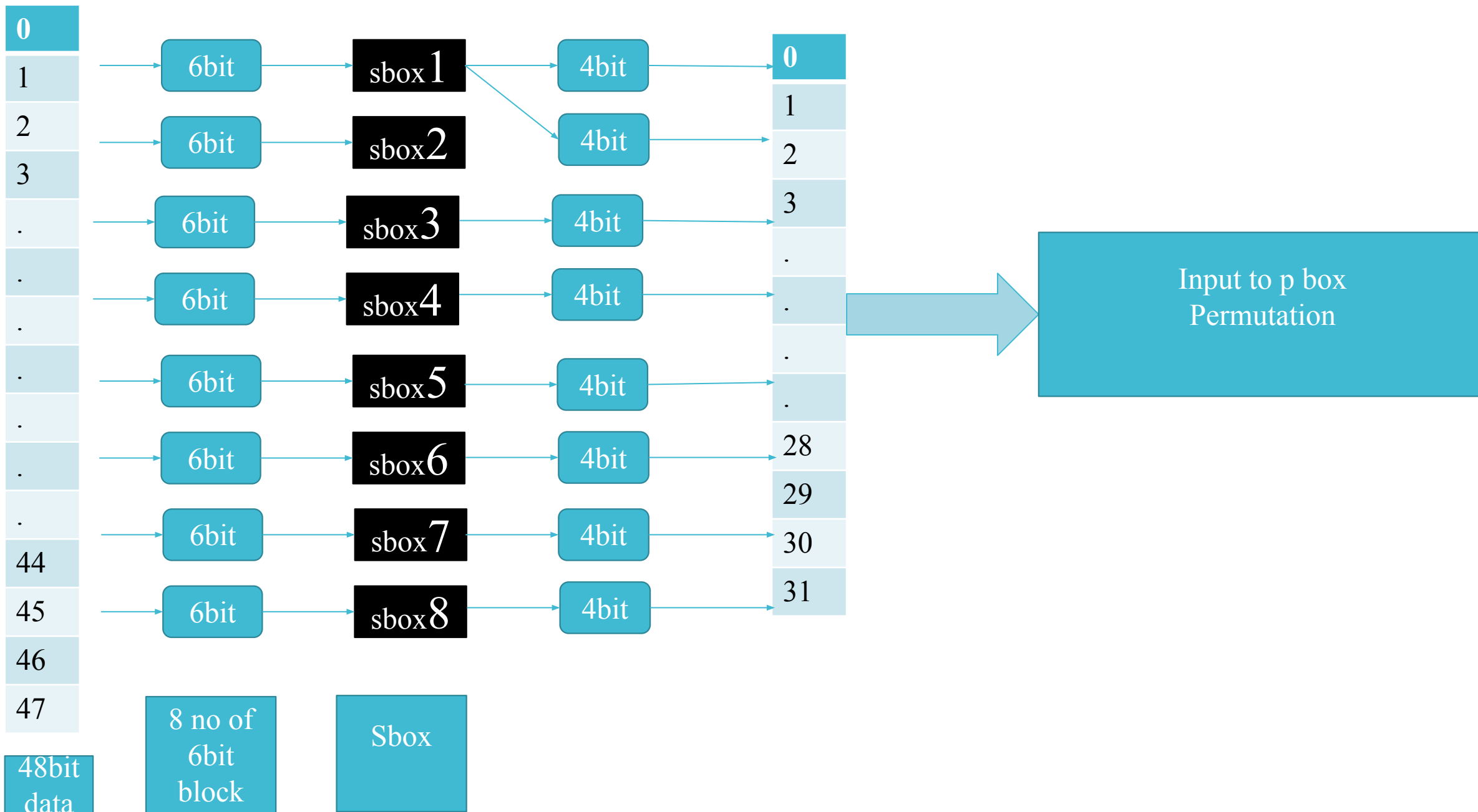


- Now we will talk about these 8 block of 4bit each in next slide.
- These 4bit block are then expand to 6 bit block in a particular manner.

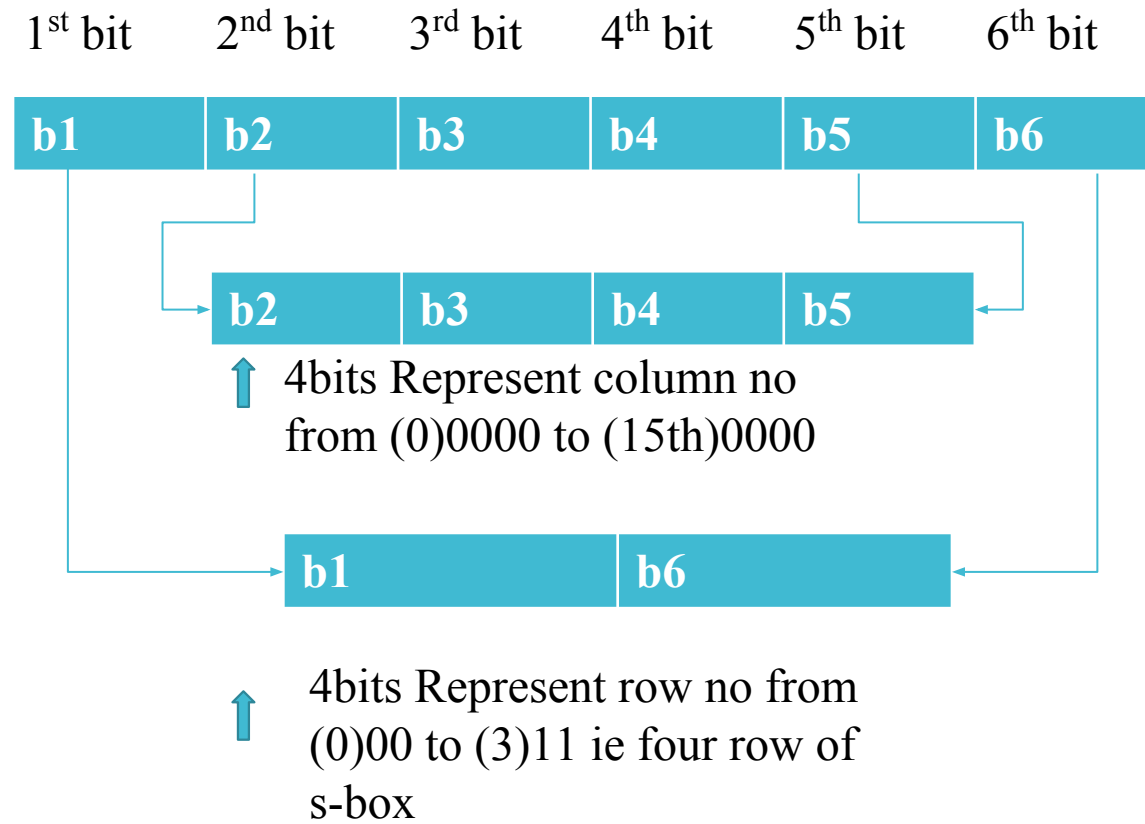


- 6bit * 8 bit = 48 bit key generation.
- Until now
- Key transformation 56 bit $\rightarrow$ 48 bits
- Expansion permutation 32bit RPT $\rightarrow$ 48bit RPT
- Now 48 bit RPT XOR 48 bit key $\rightarrow$ output $\rightarrow$ S-box substitution





How S-box works

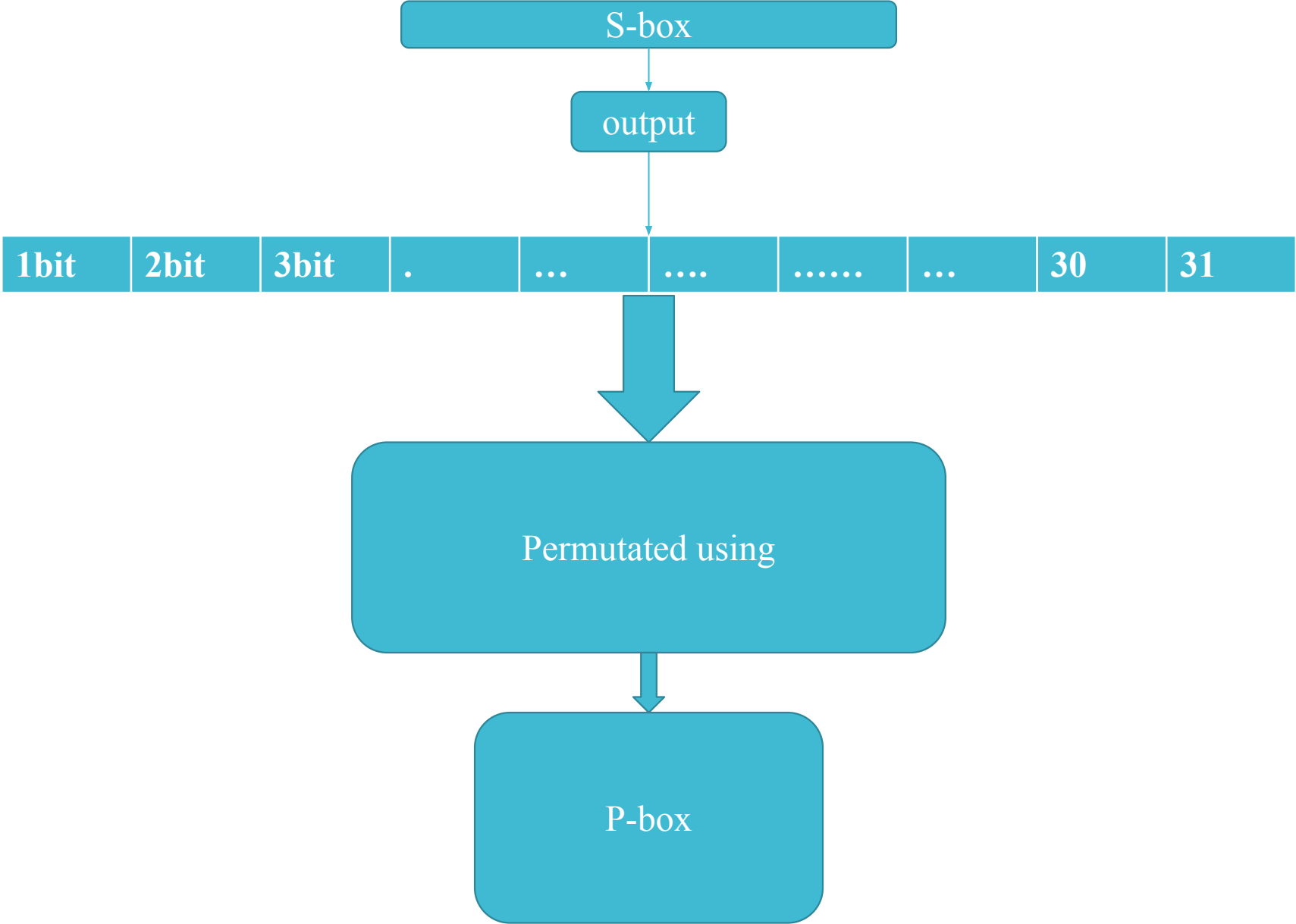


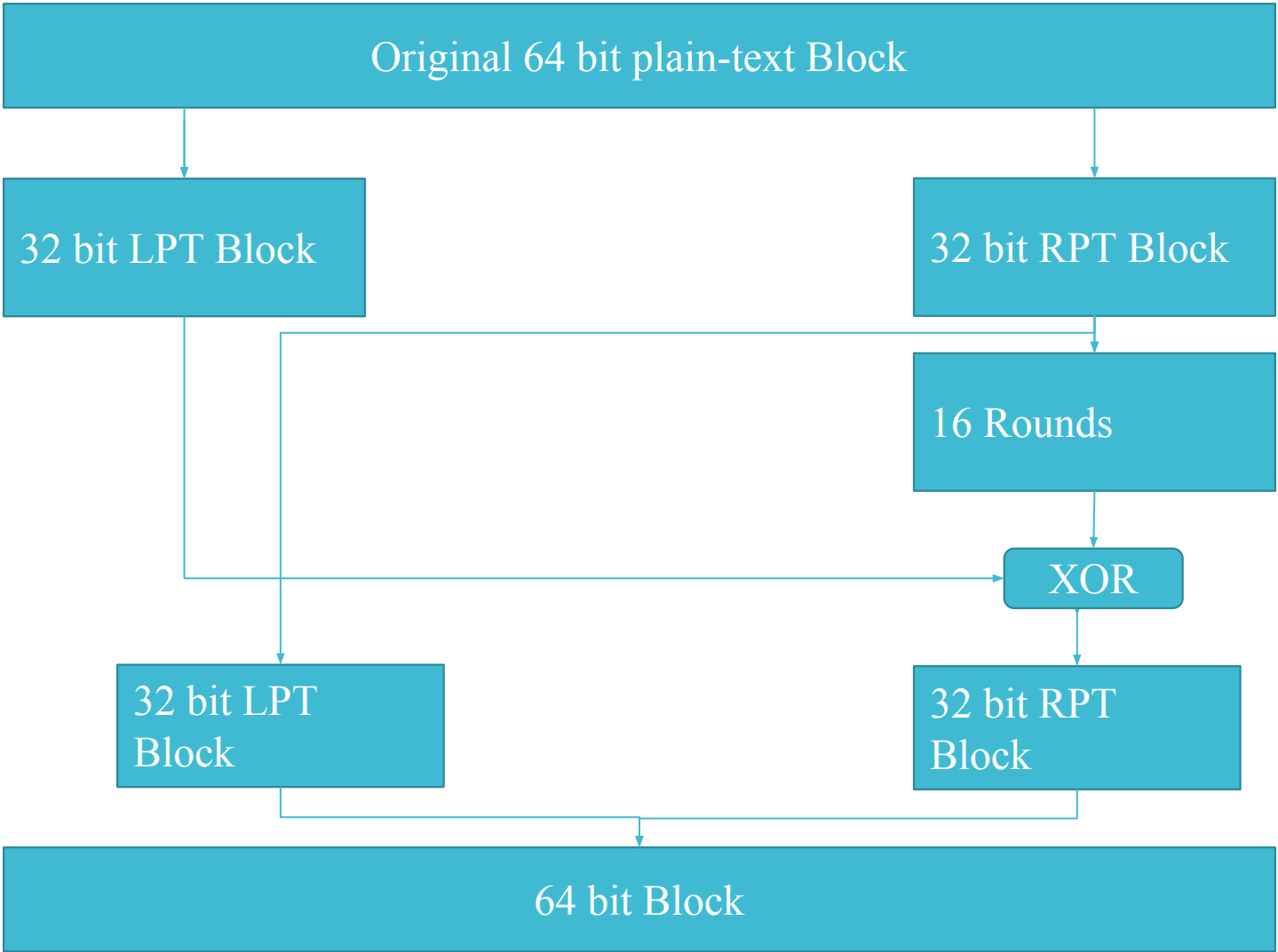
14	4	13	1	2	15	11	8	3	10	6	12	5	9	0	7
0	15	7	4	14	2	13	1	10	6	12	11	9	5	3	8
4	1	14	8	13	6	2	11	15	12	9	7	3	10	5	0
15	12	8	2	4	9	1	7	5	11	3	14	10	0	6	13

Fig. 3.32 (a) S-box 1

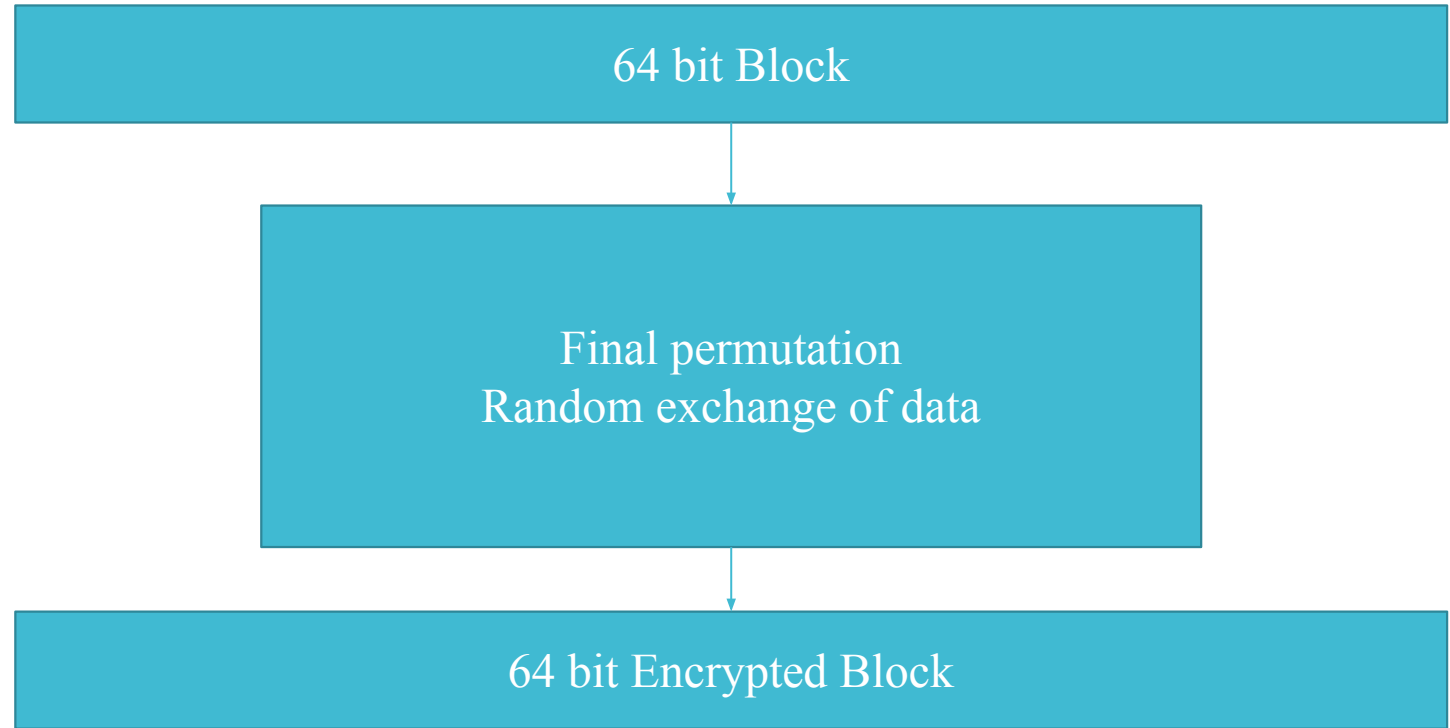
S-box 1

- Map the data in s-box located from row no bit b1b6 and column no bit b2b3b4b5
- Note* : start counting row and column from 0
- Get the data from the sbox.
- Here we are substituting the 4bit from the s-box
- Convert the no from the s box into decimal digit.
- Repeat for each s-box with each 6bit block.





Final permutation



Step 6 cipher text

- Cipher text the text that is encrypted.
- Though DES is later known as vulnerable which require the revolution of des as double and Triple des but they were also fail due to slow speed and more computation.
- Des algorithms are easily encode able with the help of cluster computing.
- However these algorithm can be used for decryption also by using the same keys generated in the 16 rounds.

Thankyou...

- This presentation is heavily inspired by the book Information and cyber security by Author Atul Kahate